CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

DIETING SYSTEM BASED ON DISEASE

Aim:

To implement a Prolog-based diet recommendation system according to a person's disease.

Objective:

To recommend suitable food items based on disease-related rules.

Algorithm:

1. Start.
2. Define disease facts.
3. Define recommended foods for each disease.
4. Query the person's disease.
5. Retrieve recommended foods.
6. Display the diet recommendation.
7. Stop.

Flowchart START

|

v

Enter Disease

|

v

Check Disease Database

|

v

Find Recommended Food

|

v

Display Diet

|

v

STOP

Prolog Program:

```
% Diet Recommendation System

disease(diabetes).
disease(hypertension).
disease(obesity).

recommended_food(diabetes, vegetables).
recommended_food(diabetes, whole_grains).
recommended_food(diabetes, low_sugar_fruits).

recommended_food(hypertension, vegetables).
```

recommended_food(hypertension, low_sodium_food).

recommended_food(hypertension, fruits).

recommended_food(obesity, vegetables).

recommended_food(obesity, salads).

recommended_food(obesity, whole_grains).

diet_for(Disease, Food)

:disease(Disease),

recommended_food(Disease,

Food).

Queries

?- diet_for(diabetes, Food).

Output:

```
85 ?- recommend_diet(diabetes, Diet).
Diet = 'Low sugar and high fiber diet'.
```

Result:

Thus, the Prolog program successfully recommends suitable food items based on the given disease.